A Project Report on

“Sales Data Analysis”

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Submitted by

Alka Kushwaha

Student

AKS University Satna

Submitted to

Namanand Maske

Founder & CEO

www.thedevelopersarena.com

PROJECT OVERVIEW

The objective of this project is to analyze a real-world sales dataset using Python and the pandas library.

The goal is to:

- Calculate total revenue
- Identify the best-selling product
- Compute statistical metrics (average, maximum, minimum sales)
- Handle missing values
- Generate a clean analysis report

This project helps in understanding real-world data processing and business insights extraction.

SETUP INSTRUCTIONS

1. Install Python (3.x)
2. Install pandas library using:
   Pip install pandas

3. Place the following files in the same folder:

- Sales_analysis.py
- Sales_data.csv
- Requirements.txt

4. Run the script:
   Python sales_analysis.py

CODE STRUCTURE

```python
# Step 1: Import required library
import pandas as pd

# Step 2: Load the dataset
df = pd.read_csv("sales_data.csv")

print("Dataset Loaded Successfully!\n")

# Step 3: Basic Information
print("First 5 Rows of Dataset:")
print(df.head())

print("\nDataset Shape (Rows, Columns):")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

# Step 4: Check Missing Values
print("\nMissing Values:")
print(df.isnull().sum())

# Fill missing values if any
df = df.fillna(0)

# Step 5: Calculate Total Revenue
total_revenue = df["Total_Sales"].sum()
print(f"\nTotal Revenue: ₹{total_revenue:,.2f}")

# Step 6: Find Best Selling Product
best_product = df.groupby("Product")["Total_Sales"].sum().idxmax()
best_product_sales = df.groupby("Product")["Total_Sales"].sum().max

print(f"\nBest Selling Product: {best_product}")
print(f"Sales of Best Product: ₹{best_product_sales:,.2f}")

# Step 7: Additional Metrics
average_sales = df["Total_Sales"].mean()
max_sale = df["Total_Sales"].max()
min_sale = df["Total_Sales"].min()

print(f"\nAverage Sale: ₹{average_sales:,.2f}")
print(f"Highest Single Sale: ₹{max_sale:,.2f}")
print(f"Lowest Single Sale: ₹{min_sale:,.2f}")

print("\nAnalysis Completed Successfully!")
```

- Import pandas library
- Load CSV file using pd.read_csv()
- Display dataset preview
- Check shape, columns, and data types
- Handle missing values using fillna()
- Perform groupby operations for product analysis
- Calculate revenue metrics
- Print formatted output

The code is clean, structured, and includes comments explaining each step.

DATA CLEANING

- Checked for missing values using: Df.isnull().sum()
- Replaced missing values with 0 using: Df.fillna(0)
- Ensured numerical calculations are correct

Analysis Performed

- Total Revenue
  Calculated by summing the Total_Sales column.

- Best-Selling Product
  Used groupby() on Product column and found product with highest total sales.

Key Findings

The dataset contains sales records of multiple products.

Total revenue gives overall business performance.

Best-selling product indicates highest demand.

Average sales show general transaction value.

Highest and lowest sales indicate range of performance.

Technical Details

Programming Language: Python

Library Used: pandas

Data Format: CSV

Techniques Used:

Data loading

Data exploration

Missing value handling

Aggregation using groupby

Statistical analysis

Testing Evidence

Script executed without errors.

Verified calculations manually.

Checked output formatting.

Confirmed best product matches highest grouped sales.

```
••• Dataset Loaded Successfully!

    First 5 Rows of Dataset:
        Product     Category  Price  Quantity  Total_Sales
    0    Laptop  Electronics   1200         5         6000
    1     Mouse  Electronics     25        20          500
    2  Keyboard  Electronics     75        10          750
    3   Monitor  Electronics    300         8         2400
    4      Desk    Furniture    150         3          450

    Dataset Shape (Rows, Columns):
    (8, 5)

    Column Names:
    Index(['Product', 'Category', 'Price', 'Quantity', 'Total_Sales'], d

    Data Types:
    Product        object
    Category       object
    Price           int64
    Quantity        int64
    Total_Sales     int64
    dtype: object

    Missing Values:
    Product        0
    Category       0
    Price          0
    Quantity       0
    Total_Sales    0
    dtype: int64

    Total Revenue: ₹11,530.00

    Best Selling Product: Laptop
    Sales of Best Product: ₹6,000.00

    Average Sale: ₹1,441.25
    Highest Single Sale: ₹6,000.00
    Lowest Single Sale: ₹200.00

    Analysis Completed Successfully!
```

Conclusion

This project demonstrates how raw sales data can be transformed into meaningful business insights using Python and pandas. It provides a strong foundation in data analysis concepts including data cleaning, aggregation, and reporting.

Through this analysis, meaningful insights were extracted from raw sales data, including revenue trends and product performance. The project also improves understanding of how data-driven decision-making supports business growth and strategic planning.

Overall, this task strengthened foundational skills in data handling, analytical thinking, and structured reporting, which are essential for future data science and analytics projects.